VISVESVARAYA TECHNOLOGICAL UNIVERSITY

Belagavi-590018

DBMS Mini Project Report on

"SCHOOL FEES MANAGEMENT SYSTEM"

Submitted to

Visvesvaraya Technological University

In the partial fulfilment of requirements for the award of degree

Bachelor of Engineering

In

COMPUTER SCIENCE AND ENGINEERING

Submitted By

KIRAN KUMAR C V	**4RA22CS405**
ASHWINI C	**4RA22CS400**

Under the Guidance of

Mrs. MONIKA M M

Assistant Professor, Dept. of CSE

Rajeev Institute of Technology,

Hassan

RAJEEV INSTITUTE OF TECHNOLOGY

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

HASSAN-573201

2023-2024

RAJEEV INSTITUTE OF TECHNOLOGY, HASSAN

(Approved by AICTE, New Delhi and Affiliated to VTU, Belagavi.)

Plot #1-D, Growth Centre, Industrial Area, B-M Bypass Road, Hassan – 573201Ph:(08172)-243180/83/84 Fax:(08172)-243183



Department of Computer Science and Engineering

CERTIFICATE

Certified that the **DBMS Mini Project** entitled **"SCHOOL FEES MANAGEMENT SYSTEM"** is carried out by **Mr. KIRAN KUMAR C V [4RA22CS405] and Ms. ASHWINI C [4RA22CS400]** respectively, a Bonafede students of **RAJEEV INSTITUTE OF TECHNOLOGY**, Hassan in partial fulfilment for the award of **BACHELOR OF ENGINEERING** in **COMPUTER SCIENCE AND ENGINEERING** of the Visvesvaraya Technological University, Belagavi during the year 2023-2024. The Mini Project Phase-1 report has been approved as it satisfies the academic requirements in respect of mini project work prescribed for the said Degree.

Mrs. MONIKA M M	**Dr. SHRISHAIL MATH**	**Dr. MAHESH P K**
Assistant Professor	Head of the Department	Principal
Dept. of Computer Science & Engineering	Computer Science & Engineering	**RIT, HASSAN**
RIT, HASSAN	**RIT, HASSAN**	

Name of the examiners	Signature with date
1.	.....
2.	.....

DECLARATION

We **KIRAN KUMAR C V(4RA22CS405), ASHWINI C(4RA22CS400)** bearing students of 5th Sem B.E in **Computer Science and Engineering**, **Rajeev Institute of Technology, Hassan**, hereby declare that the work being presented in the dissertation entitled " **SCHOOL FEES MANAGEMENT SYSTEM "** has been carried out by us under the supervision of guide **Mrs. Monika M M**, Assistant Professor, Computer Science and Engineering, **Rajeev Institute of Technology, Hassan**, as partial fulfilment of requirement for the award of B.E Degree of **Bachelor of Engineering in Computer Science and Engineering at Visvesvaraya Technological University, Belagavi** is and authenticrecord of my own carried out by us during the academic year 2023-2024.

KIRAN KUMAR C V
4RA22CS405

ASHWINI C
4RA22CS400

PLACE : HASSAN
DATE : .....

ACKNOWLEDGEMENT

The satisfaction and euphoria that accompany the successful of any task would be incomplete without the mention of the people who made it possible, whose constant guidance and encouragement crowned our efforts with success.

We would like to profoundly thank our **Management of Rajeev Institute of Technology** & our President **Dr. Rachana Rajeev** for providing such a healthy environment.

We would like to express our sincere thanks to our principal **Dr. Mahesh P K**, Rajeev Institute of Technology for his encouragement that motivated us for successful completion of mini project work.

We wish to express our gratitude to **Dr. Shrishail Math**, Head of the Department of Computer Science& Engineering for providing a good working environment and for his constant support and encouragement.

It gives us great pleasure to express our gratitude to **Mrs. Monika M M** Assistant Professor, Department of Computer Science & Engineering for her expert guidance, initiative and encouragement that led us to complete this mini project.

We would also like to thank all our staffs of Computer Science and Engineering department who have directly or indirectly helped us in the successful completion of this phase-1 project and, we would like to thank our parents.

KIRAN KUMAR C V
4RA22CS405

ASHWINI C
4RA22CS400

ABSTRACT

This project “School Fees Management System” is a desktop system enables efficient storage of student records to properly manage the fee records of the students. And it also generates messages for due balances its student’s fee. The system is designed for fee management of a college administration department. It makes searching records easier and faster.

Furthermore, the system may include functionalities such as:

1. User authentication and authorization to ensure data security and privacy.
2. Reporting features to generate comprehensive reports on fee collections, outstanding balances, etc.
3. Integration with payment gateways for online fee payments.
4. Customizable settings to accommodate the specific needs and policies of the institution.
5. Automated calculations for fees, including tuition, exam fees, library dues, etc.
6. Backup and recovery mechanisms to prevent data loss.

Contents

DECLARATION i

ACKNOWLEDGMENT ii

ABSTRACT iii

TABLE OF CONTENT iv

LIST OF SNAP SHOTS v

CHAPTER 1 **PAGE NO**

1.INTRODUCTION

1.1 Introduction to database 1

1.2 Introduction 3

1.3 Objective 3

1.4 About system 3

1.5 Scope of the system 4

1.6 Existing System and drawback 4

1.7 Proposed System & Advantages 5

2. FEASIBILITY STUDY

2.1 Economical feasibility 6

2.2 Technical feasibility 6

2.3 Operational feasibility 6

3. REQUIREMENTS SPECIFICATION

3.1 Hardware requirements 8

3.2 Software requirements 8

3.3 Description of front end 8

3.3.1 Hypertext mark-up language 8

3.3.2 Hypertext mark-up language 8

3.4 MySQL 11

4. SYSTEM DESIGN

4.1 Entity relationship diagram 12

4.2 Schema Diagram 13

4.3 Table description 14

5. SNAPSHOTS 18

6. CONCLUSION

6.1 Conclusion 23

7. REFERENCES

7.1 References 24

LIST OF SNAPSHOTS

Table No.	Name of Table.	Page No.
4.3.1	Course	14
4.3.2	Fees	14
4.3.3	Payment	15
4.3.4	Student	15
4.3.5	Student of list	16
4.3.6	System setting	16
4.3.7	Users	17

Snapshot No.	Snapshot Name.	Page No.
5.1	Login Page	18
5.2	Dashboard	18
5.3	Student fees	19
5.4	Payments	19
5.5	Courses & Fees	20
5.6	Students	20
5.7	Payments Report	21
5.8	Users	22

CHAPTER 1

INTRODUCTION

1.1 Introduction to database

A database is an organized collection of data. A relational database, more restrictively, is a collection of schemas, tables, queries, reports, views, and other elements. Database designers typically organize the data to model aspects of reality in a way that supports process requiring information, such as (for example) modelling the availability of rooms in hotels in a way that supports finding a hotel with vacancies.

A database-management system (DBMS) is a computer-software application that interacts with end-users, other applications, and the database itself to capture and analyse data. A general-purpose DBMS allows the definition, creation, querying, update, and administration of databases. Well-known DBMSs includes MySQL Microsoft SQL Server, Oracle, Sybase, SAP HANA, MySQL, SQLite and IBM DB2.

A database is not generally portable across different DBMSs, but different DBMSs can interoperate by using standards such as SQL and ODBC or JDBC to allow a single application to work with more than one DBMS. Computer scientists may classify database-management systems according to the database models that they support; the most popular (database systems since the 1980s have all supported the relational model generally associated with the SQL language. Sometimes a DBMS is loosely referred to as a "database".

The technology progress in the areas of processors, computer memory, computer storage, and computer networks, the sizes, capabilities, and performance of databases and their respective DBMSs have grown in orders of magnitude. The development of database technology can be divided into three eras based on data model or structure: navigational, SQL/relational, and post-relational. The two main early navigational data models were the hierarchical model, epitomized by IBM's IMS system, and the CODASYL model (network model), implemented in a number of products such as IDMS.

Databases are fundamental components of modern information systems and are widely used in business, education, research, and many other fields. Understanding these fundamental concepts will provide a solid foundation for working with databases and developing applications that rely on them.

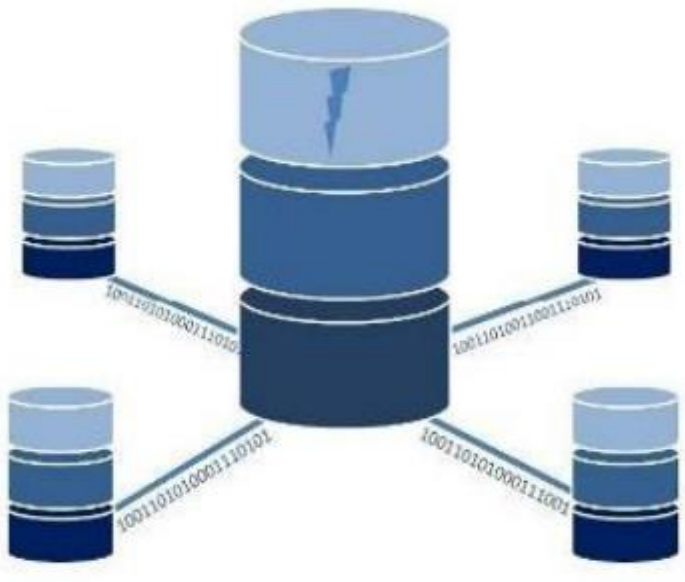


Fig1.1: Sample figure of database

The relational model, first proposed in 1970 by Edgar F. Codd, departed from this tradition by insisting that applications should search for data by content, rather than by following links. The relational model employs set of ledger-style tables, each used for a different type of entity. Only in the mid-1980s did computing hardware become powerful enough to allow the wide deployment of relational systems (DBMSs plus applications). By the early 1990s, however, relational systems dominated in all large-scale data processing applications, and as of 2015 they remain dominant: IBM DB2, Oracle, MySQL, and Microsoft SQL Server are the top DBMS.

The dominant database language, standardized SQL for the relational model, has influenced database languages for other data models. Object databases were developed in the 1980s to overcome the inconvenience of object relational impedance mismatch, which led to the coining of the term "post-relational" and also the development of hybrid object-relational databases.

The next generation of post-relational databases in the late 2000s became known as MySQL databases, introducing fast key-value stores and document-oriented databases. A competing "next generation" known as MYSQL databases attempted new implementations.

1.2 Introduction

This project "**School Fees management System**" is a desktop system enables efficient storage of student records to properly manage the fee records of the students. And it also generates messages for due balances its student's fee.

The system is designed for fee management of a college administration department. It makes searching records easier and faster.

Fees Management System project is developed using PHP, CSS, and JavaScript. Talking about the project, it has all the essential features. This project has an administration side from Where he/she can view branch, students, fees, report, manage fees, students, branches, settings. In this project, all the functions are performed from the admin side which means there is no user side.

1.3 Objective:

- To reduce paperwork
- To make storage of information more efficient and secure.
- To have a friendly interface.
- To operate it easily and with minimum experience.
- To save time and energy of the admin.

1.4 About system:

Admin has full control of the system, he/she can view branch, students, fees and manage branch, students, fees from the system. The project also includes a Fees report of students in Report module, which displays Fees Information as well as respective Student information. He/she can add, edit, delete, view Branch. While adding Branch, he/she has to provide Branch Name, Address, and Detail. Likewise, while adding students, he/she has to provide Personal information like Name, Contact, branch, DOJ, Fees Information like Total Fees, Advance Fee, Remarks and Optional Information like About student and Email id. To take Fees for a student, the user has to provide Paid amount, date and Remarks. After paying fees of the student, that particular name will be removed from Fees module.

1.5 Scope of the system:

• Specifically designed for an individual college.

• Inserting new student records are not possible.

• It is based on desktop application.

• Not including fees other than academic like bus fees and etc.

• There are only limited number of modules for fees management

1.6 Existing System and drawback:

• In the existing system, colleges have to manually maintain information regarding to Fees deposited by the students.

• College management system is complex and time consuming to maintain fees of students by that very difficult.

• It is not properly capable to manage the student records with their fee details at a single place.

• Managing collection of student fees, issuing fee receipts and fee register updating is a laborious manual process, leading to data inaccuracy and / or reconciliation.

• Preparing receipts manually everyday needs additional clerical staff.

• To generate due fees report is required a complete manual procedure, which involves a lot of time and clerical staff manpower.

• Re-entry of fees receipts in accounting software separately leads to double manpower cost and time.

1.7 Proposed System & Advantages:

• The fees management System is a desktop system aimed to maintaining students records and their fees details.

• It also generates records like I.e. Feed Paid, dues, and etc.

• The system requires small amount of time to generate reports needed to manage the fees of the student.

- Managing collection of school fees, issuing fee receipts and fee register updating is done with the help of software resulting in highly accurate data.
- Software provides facility to print receipts automizing office work.
- This Software provides facility to generate due fees report easily and at any point of time.
- Software directly enters fees receipts to the accounts of the school.

CHAPTER 2

FEASIBILITY STUDY

After doing the project college online traffic offense management system, study and analysing all the existing or required functionalities of the system, the next task is to do the feasible study for the project. All projects are feasible-given unlimited resources and infinite time.

Feasibility study included consideration of all the possible ways to provide a solution to the given problem. The proposed system should satisfy all the user requirements and should be feasible enough so that future changes can be easily done based on the future upcoming requirements.

2.1 Economical feasibility:

This is very important aspect to be considered while developing project we decided the technology based on minimum cost factor. All hardware and software cost as to be borne by the institution.

- Overall, we have estimated that the benefits the organization is going to receive from the proposed system will surely overcome the initial cost and the later on running cost for the system.

2.2 Technical feasibility:

This includes the study of function, performance and constraints that may affects the ability to achieve an acceptable system. For this feasibility study, we studied complete functionality to be provided in the system, has described in the system requirement specification, and checked if everything was possible in different type of front end and backend platform.

2.3 Operational feasibility:

No doubts the proposed system is fully user friendly and all inputs to be taken all self-explanatory even to a layman. As far our study is concerned the clients are comfortable and happy as the system has cut down their loads and doing. Some of the important issues raised are to test the operational feasibility of a project includes the following:

- Is there sufficient support for the management from the users?

- Will the system be used and work properly if it is being developed and implemented?

- Will there be any resistance from the user that will undermine the possible application benefits? This system is targeted to be in accordance with the above-mentioned issues. Beforehand, the management issues and user requirements have been taken into consideration. So, there is no question of resistance from the users that can undermine the possible application benefits. The well-planned design would ensure the optimal utilization of the computer resources and would help in the improvement of performance status.

CHAPTER 3

REQUIREMENTS SPECIFICATION

3.1 Hardware requirements

- Processor : PENTIUM or any 64-bit processor
- Processor Speed : 1.6 GHz or above
- RAM : 2GB or above
- Hard Disk : 20GB or above

3.2 Software requirements

- Operating System : Windows 8,10 or 11/Ubuntu 18 or above
- Front end : HTML,CSS, JavaScript
- Back end : MYSQL,PHP
- Server : Apache server

3.3 Description of front end

3.3.1 Hypertext mark-up language

The Hyper Text Mark-up language (HTML) is a simple mark-up language used to create hypertext documents that are portable from one platform to another HTML documents are SGML documents with generic semantics that are appropriate for representing information from a wide range of applications. This specification defines HTML version 4.0 HTML 4.01 aims to capture recommended practice as of early'96 and as such to be used as a replacementforHTML3.2

Why to use HTML

Website is collection of pages, publications and documentation that reside on webserver. While these page publication and a document as a formatted in any single format you should use HTML for home page and all primary pages and the site. This will enable the millions of web users it considered first for matting any new material, you plant publish on the web HTML documents are platform independent, meaning that they don't conform to any standard it they are created properly you can more home page to any server platform or you can access them with any complaint www.browser.

- <HTML>…</HTML>

-All HTML files start and end with the tag pair.

- <HEAD>...</HEAD>

- All HTML have a pair of --HEADI tags that indicate what the tile and other attributes of the page are going to be.

- <TITLE>...<TITLE>

-This tag indicates what the title of the HTML file is going to be on the BROWSER window

title.

- <BODY>...</BODY>

-This tag pair is to logically separate the HTML file into the header and the body. Usually, the header contains information regarding Training & Placement System the html where as the body contains information that the HTML file must actually contain.

- The HTML template must look like.

<!DOCTYPEHTMLPUBLIC-THIS IS ANEXAMPLE>

<HTML><HEAD><TITLE> YOUR TITLE GOES

HERE</TITLE></HEAD></HTML>

- <P>...</P>

ss- This tag pair used to indicate the paragraph. Any text that needs to be separated into a paragraph must be put in within a paragraph tag.

- <B>...</B>

-This tag pair is used to indicate the text within tag pair must be in bold letters.

- <I>...</I>

-This tag pair is used to indicate the text with in the tag pair must be in italic letters.

- <HIALIGN=CENTER>...</HI>

-This pair of tags is used to indicate that the text must be main title for the HTML page. The ALIGN attribute can be used to set the alignment to centre or left or right

- <H1>Heading1</H1>
- <H2>Heading2</H2><H3>Heading3</H3>

-This set of tags will show the Headings in smaller fonts as the heading increases.

- ALIGN

-The align attribute can be used for headings as well. For <P>...</P> tags also, the ALIGN

attribute can be used.

-

- Used to insert a carriage return in the HTML file. The attribute to be used for this is the CLEAR attribute.

- <CENTER>...</CENTER>

-To centre the entire block of text these tags are used.

- <A>...</A>

- Anchor Tags. These tags are used linking namely hyperlinking.

3.3.2 Features of PHP

PHP is a general-purpose server-side scripting language originally designed for Web development to produce dynamic Web pages. It is one of the first developed server-side scripting languages to be embedded into an HTML source document rather than calling an external file to process data. The codes interpreted by a Web server with a PHP processor module which generates the resulting Web page. It also has evolved to include a command-line interface capability and can be used in standalone graphical applications. PHP can be deployed on most Web servers and also as a stand-alone shell on almost every operating system and platform free of charge. A competitor to Microsoft's Active Server Pages (ASP) server-side script engine and similar languages, PHP is installed on more than 20million Web sites and 1 million Webservers. Software that uses PHP includes Joomla, Word press, My BB, and

Drupal. PHP was originally created by Ramus Leadoff in 1995. The main implementation of PHP is now produced by The PHP Group and serves as the formal reference to the PHP language. PHP is free software released under the PHP License, which is incompatible with the GNU General Public License (GPL) due to restrictions on the usage of the term PHP While PHP originally stood for "Personal Home Page", it is now said to stand for "PHP: Hypertext Pre-processor", a recursive acronym

Description of back end

3.4 MySQL

MySQL is an object relational data base management system. It offers capabilities of both relational and object-oriented database system. In general object can be defined as reasonable software codes, which are location independent and perform a specific task on any application environment with little or no change.

Oracle products are based on a concept knows are client's server technology. This concept involves segregating the processing of the application between two systems. One performs all the activity related to database and the other performs the activity to interact with the application. A client or a font end data base application also interact with the database by requesting the user and the database further it also checks four validations against the data enter by the user. The commonly used front-end tools of MySQL are MySQL-Browser.

CHAPTER 4

SYSTEM DESIGN

4.1 Entity relationship diagram:

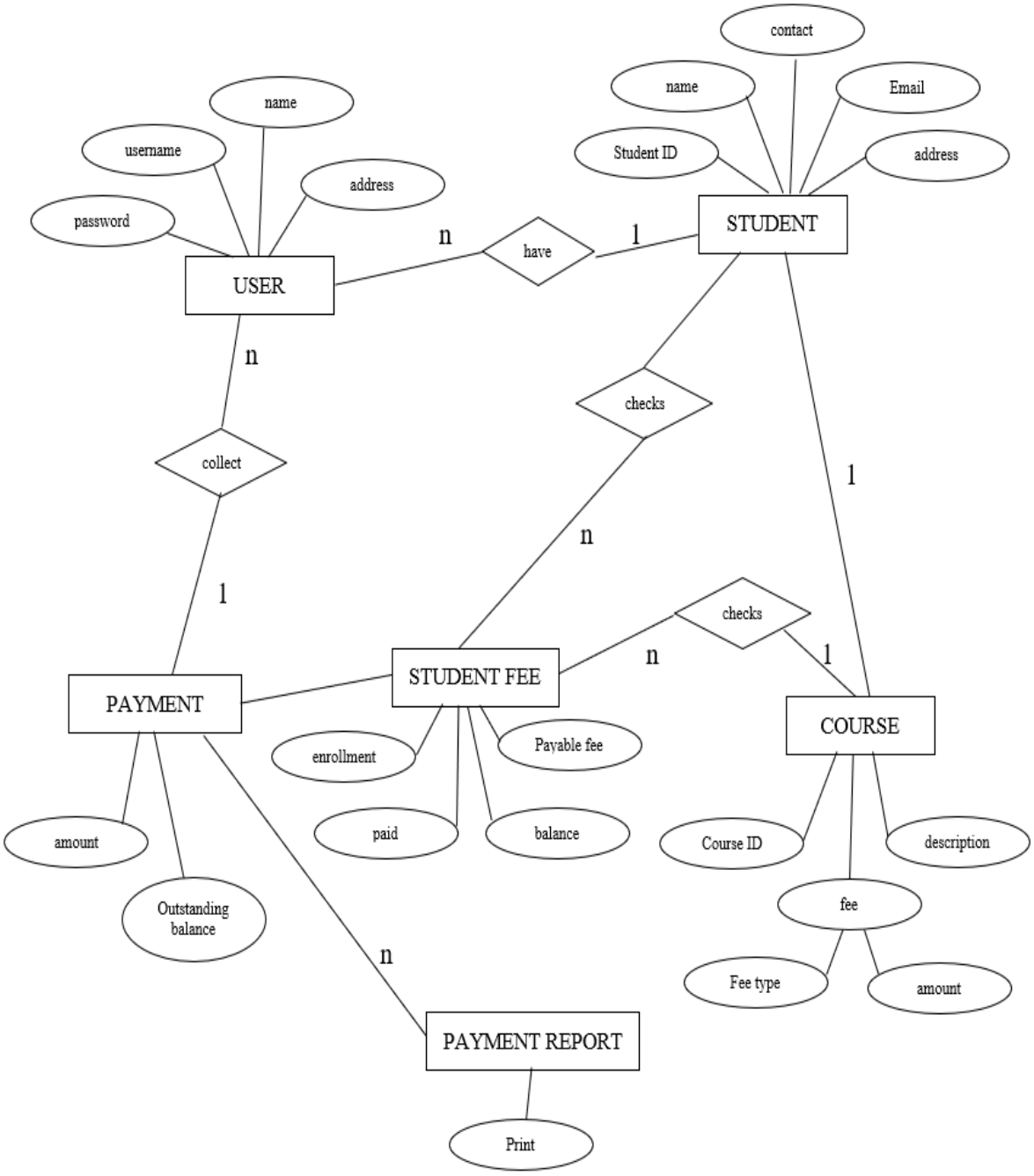
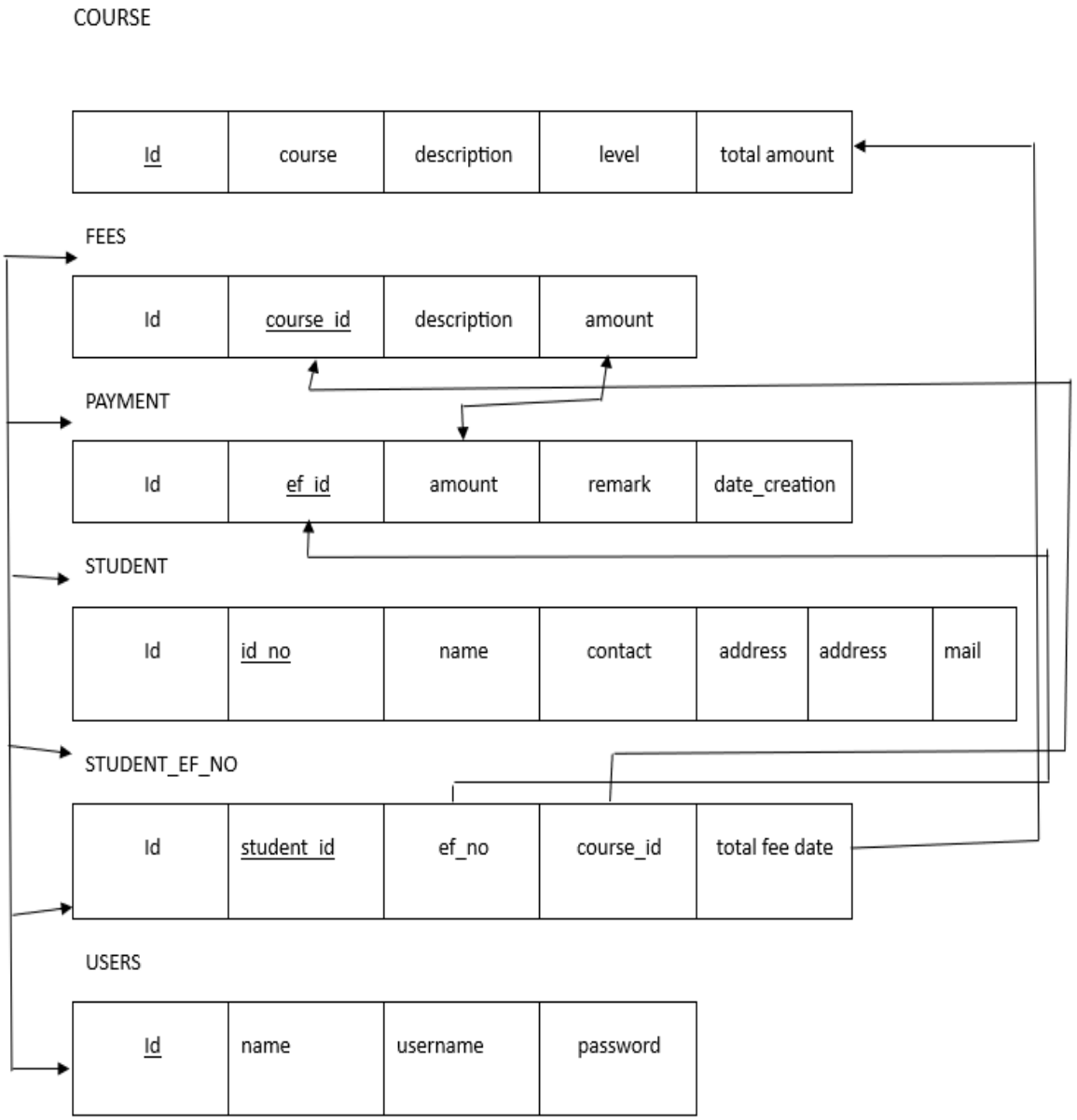


Fig4.1: Entity relationship diagram

An entity relationship diagram (ERD)shows the relationships of entity sets stored in a database. An entity in this context is a company of data. In other words, ER diagrams illustrate the logical structure of databases. At first glance entity relationship diagram looks very much like a flowchart. It is the specialized symbols, and the meanings of these symbols, that make it unique. Entities may be characterized not may be relationships, but also by additional properties (attributes), which include identifiers called “primary keys”.

4.2 Schema Diagram:



4.3 Table description:

Table Name: Course:

Primary key: id

It contains all the details of the course. It contains attributes such as course, description, level, total amount, date created.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id	int(30)			No	None		AUTO_INCREMENT
2	course	varchar(100)	utf8mb4_general_ci		No	None		
3	description	text	utf8mb4_general_ci		No	None		
4	level	varchar(150)	utf8mb4_general_ci		No	None		
5	total_amount	float			No	None		
6	date_created	datetime			No	current_timestamp()		

Table 4.3.1: Course

Table Name: Fees:

Primary key: id

It contains all the details of the category list. It contains attributes such as course_id, description, amount.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id	int(30)			No	None		AUTO_INCREMENT
2	course_id	int(30)			No	None		
3	description	varchar(200)	utf8mb4_general_ci		No	None		
4	amount	float			No	None		

Table 4.3.2: Fees

Table Name: Payment:

Primary key: id

It contains all the details of the payment. It contains attributes such as ef_id, amount, remarks.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id	int(30)			No	*None*		AUTO_INCREMENT
2	ef_id	int(30)			No	*None*		
3	amount	float			No	*None*		
4	remarks	text	utf8mb4_general_ci		No	*None*		
5	date_created	datetime			No	current_timestamp()		ON UPDATE CURRENT_TIMESTAMP(

Table 4.3.3: Payment

Table Name: Student:

Primary key: id

It contains details of student.it contains attributes such as id_no, name, contact, address, email, date_created.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id	int(30)			No	*None*		AUTO_INCREMENT
2	id_no	varchar(100)	utf8mb4_general_ci		No	*None*		
3	name	text	utf8mb4_general_ci		No	*None*		
4	contact	varchar(100)	utf8mb4_general_ci		No	*None*		
5	address	text	utf8mb4_general_ci		No	*None*		
6	email	varchar(200)	utf8mb4_general_ci		No	*None*		
7	date_created	datetime			No	current_timestamp()		

Table 4.3.4: Student

Table Name: Student_of_list:

Primary key: id

it contains details of Student_of_list. It contains attributes such as student_id, ef_no, course_id, total_fee, date_created.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id	int(30)			No	*None*		AUTO_INCREMEN
2	student_id	int(30)			No	*None*		
3	ef_no	varchar(200)	utf8mb4_general_ci		No	*None*		
4	course_id	int(30)			No	*None*		
5	total_fee	float			No	*None*		
6	date_created	datetime			No	current_timestamp()		

Table 4.3.5: Student of list

Table Name: System settings:

Primary key: id

it contains details of System settings. It contains attributes such as name, email, contact, cover_img, about_content.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	id	int(30)			No	*None*		AUTO_INCREMEN
2	name	text	utf8mb4_general_ci		No	*None*		
3	email	varchar(200)	utf8mb4_general_ci		No	*None*		
4	contact	varchar(20)	utf8mb4_general_ci		No	*None*		
5	cover_img	text	utf8mb4_general_ci		No	*None*		
6	about_content	text	utf8mb4_general_ci		No	*None*		

Table 4.3.6: System setting

Table Name: Users:

Primary key: id

it contains details of Users. It contains attributes such as name, username, password, type.

#	Name	Type	Collation	Attributes	Null	Default	Comments	Extra
1	**id**	int(30)			No	*None*		AUTO_INCREMENT
2	**name**	text	utf8mb4_general_ci		No	*None*		
3	**username**	varchar(200)	utf8mb4_general_ci		No	*None*		
4	**password**	text	utf8mb4_general_ci		No	*None*		
5	**type**	tinyint(1)			No	3	1=Admin,2=Staff	

Table 4.3.7: Users

CHAPTER 5:

SNAPSHOTS

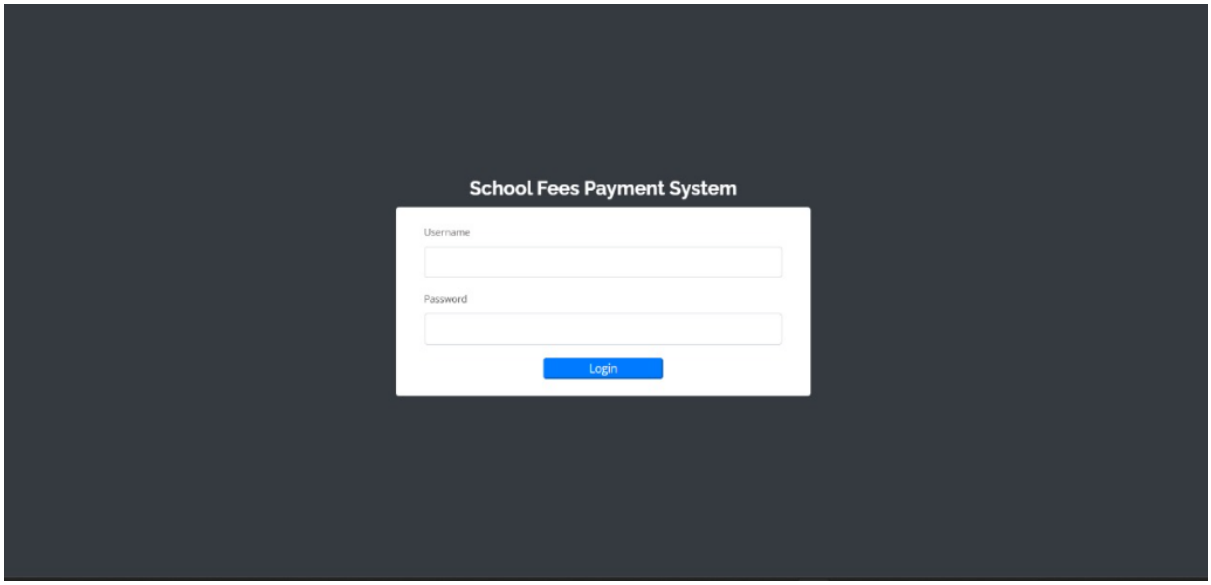


Fig.5.1: Login Page

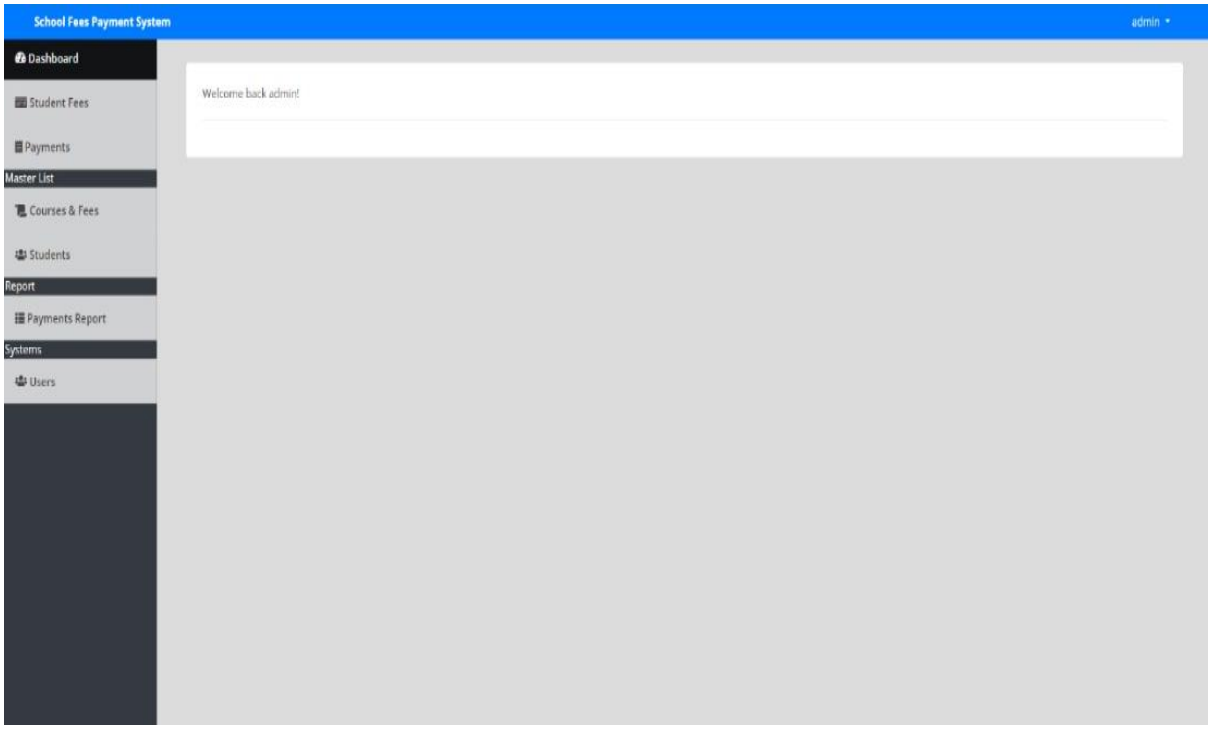


Fig.5.2: Dashboard

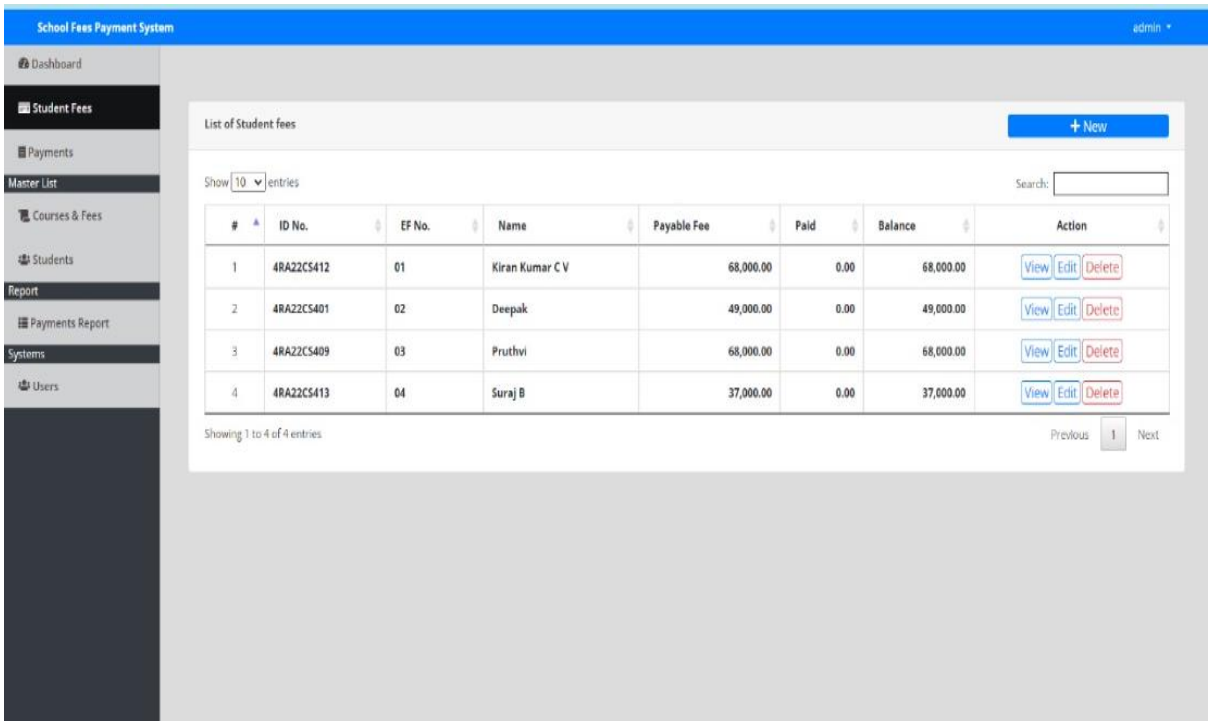


Fig.5.3: Student fees

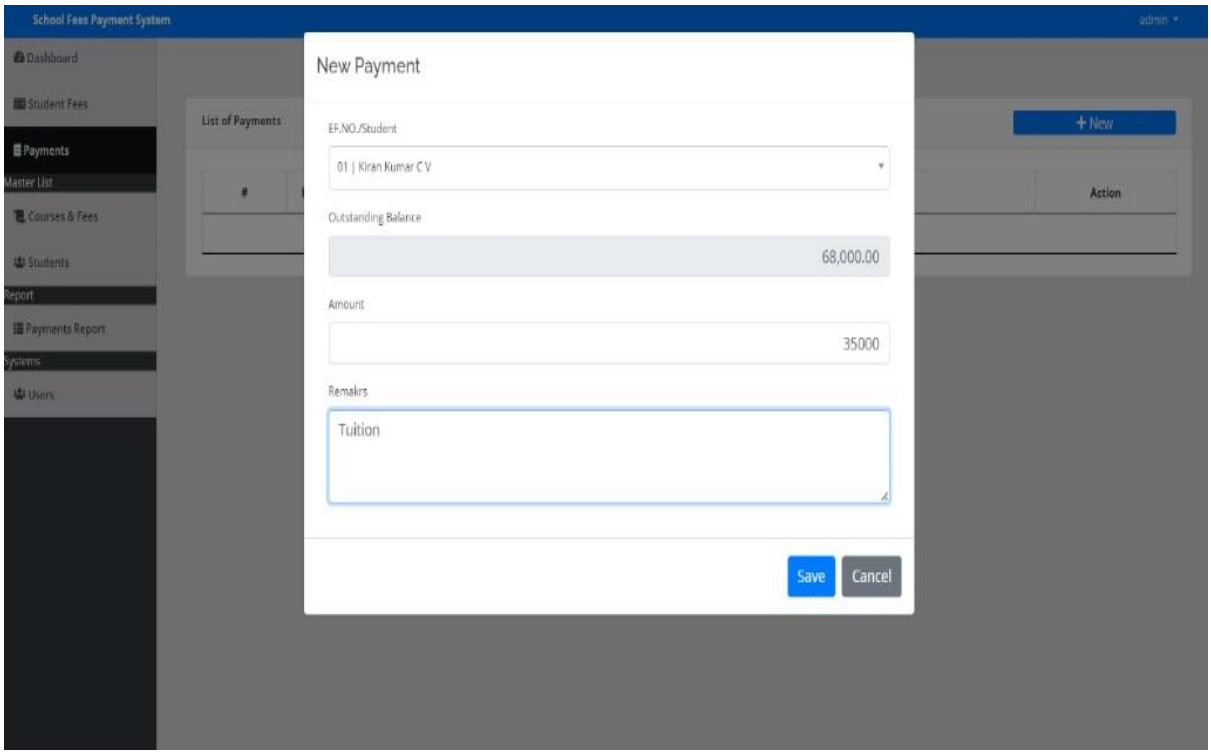


Fig.5.4: Payments

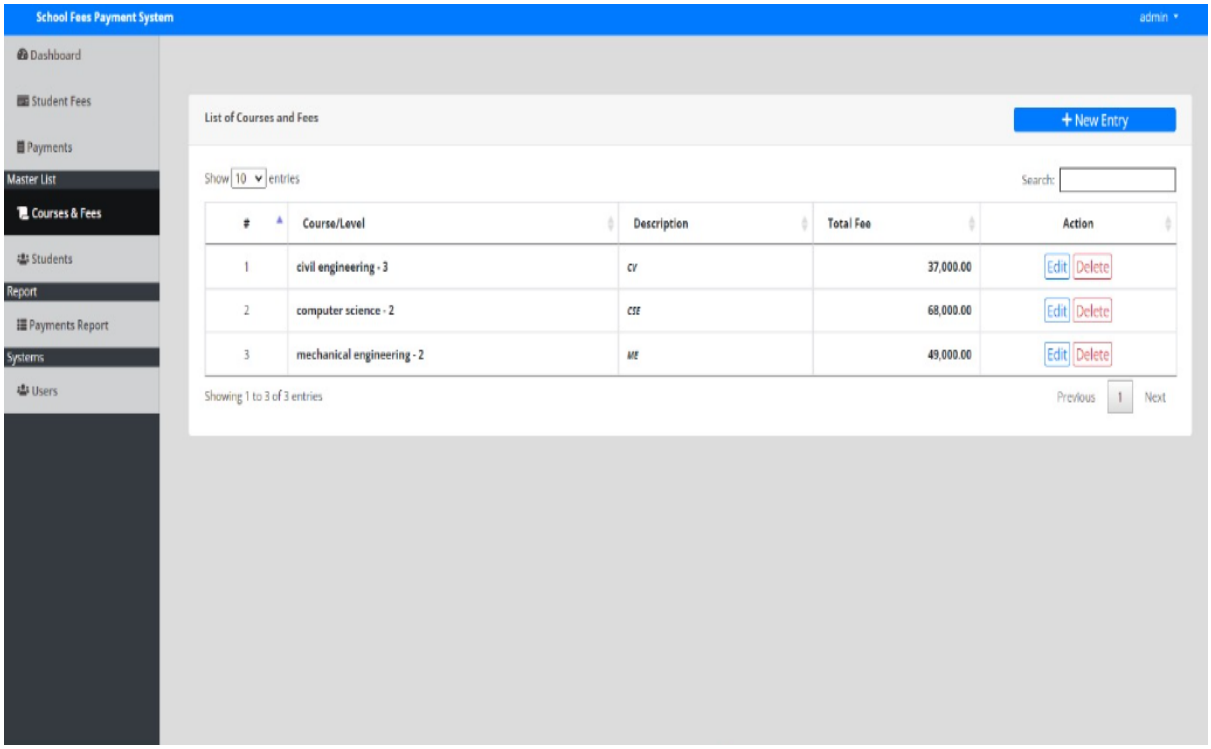


Fig.5.5: Courses & Fees

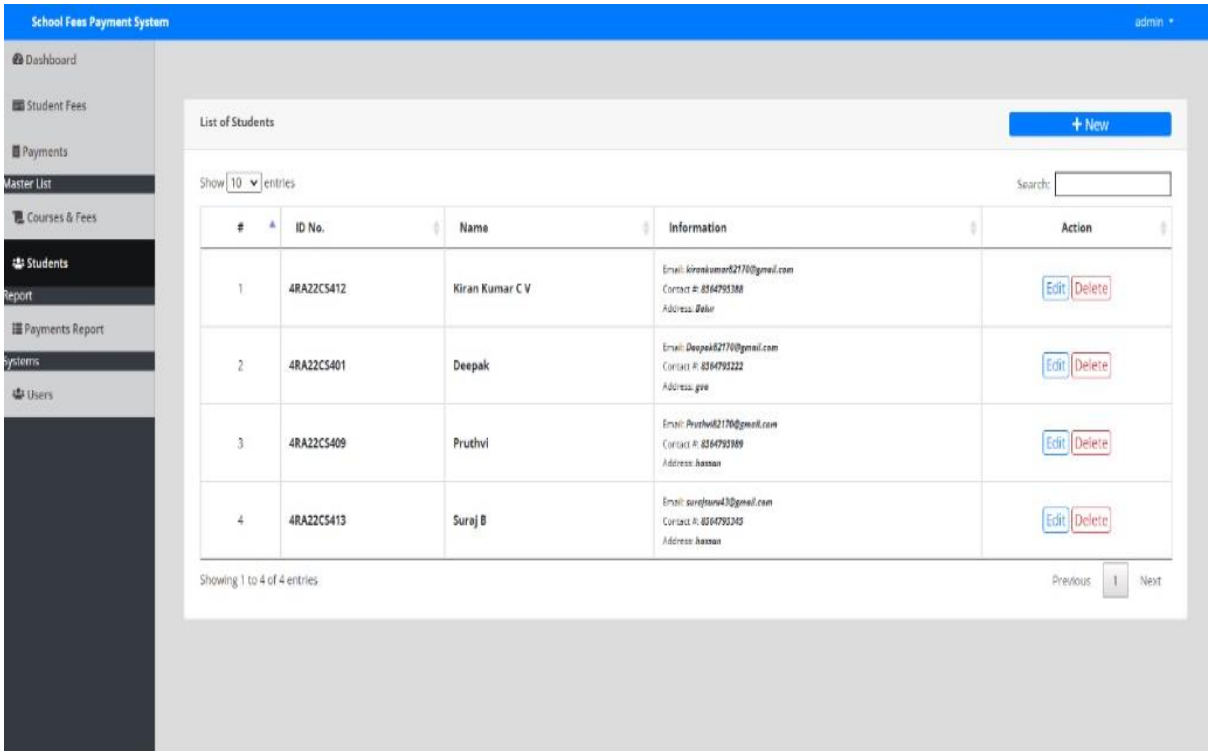
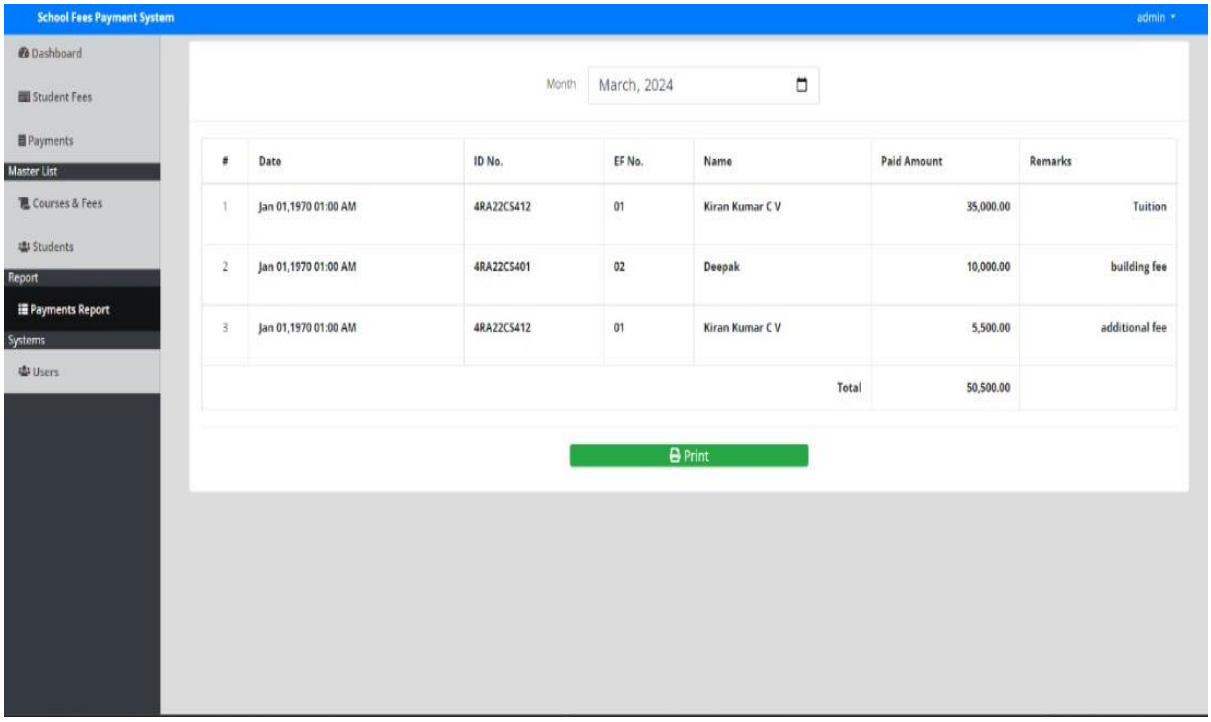


Fig.5.6: Students



Payment Receipt

EF. No: **01**
Student: **Kiran Kumar C V**
Course/Level: **computer science - 2**

Payment Date: **Mar 14,2024**
Paid Amount: **35,000.00**
Remarks: **Tuition**

Payment Summary

Fee Details	
Fee Type	Amount
tuition	**50,000**
Internship	**5,000**
building fee	**10,000**
sports	**1,000**
Additional fee	**2,000**
Total	**68,000**

Payment Details	
Date	Amount
2024-03-14	**35,000**
Total	**35,000**
Total Payable Fee	**68,000**
Total Paid	**35,000**
Balance	**33,000**

Fig.5.7: Payments Report

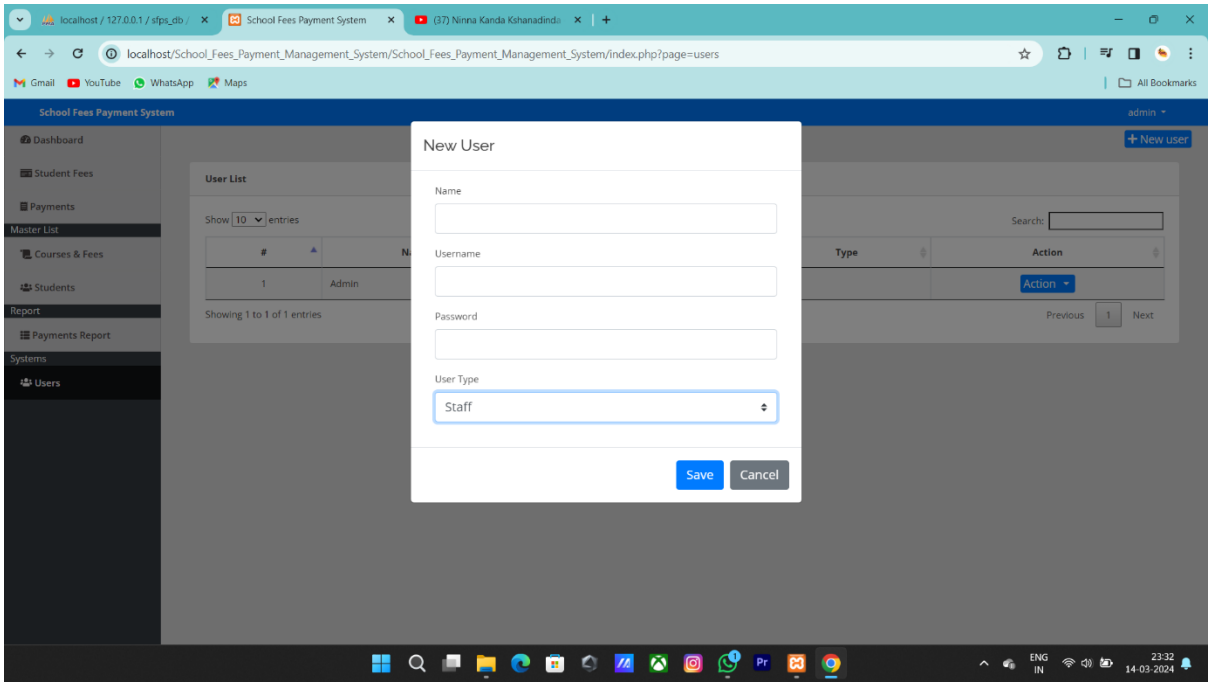

School Fees Payment System
admin
Dashboard
Student Fees
Payments
Master List
Courses & Fees
Students
Report
Payments Report
Systems
Users
New user
User List
Show 10 entries
Search:
Name Type Action
1 Admin Action
Showing 1 to 1 of 1 entries
Previous 1 Next
New User
Name
Username
Password
User Type
Staff
Save
Cancel

Fig.5.8: Users

CONCLUSION

The project of “School Fee Management System” is designed in order to reduce the maintaining bulk of records of all student fees details of who study in an Educational Institution. Inserting, retrieving and updating the fees details of a student are easy when it is compared to the manual register and storing. Maintaining the project is also easy which can be easily understandable. Maintaining the database is manageable.

In conclusion, implementing a school fees management system offers numerous benefits such as streamlined fee collection, improved financial transparency, and enhanced administrative efficiency. However, it also presents several challenges including initial costs, maintenance requirements, technical issues, data security concerns, and the need for staff training. Despite these challenges, with careful planning, investment in appropriate technology and training, and ongoing support, schools can effectively overcome these hurdles and leverage the advantages of a fee management system to better serve their students, parents, and staff while ensuring financial stability and accountability.

REFERENCES

Books referred during this project work:

- Database Management System
- System Analysis and Design.

Website referred:

- http://github.com
- http://google.com
- www.mysql.com
- www.google.co.in
- www.w3schools.com
- www.studytonight.com
- www.beginneres.com